Experiment 3

Question

Perform basic image handling and processing operations on an image. Read an image in Python and convert the image to show outlines using the Canny function.

Aim

To read an image using Python and OpenCV and detect its edges (outlines) using the Canny Edge Detection algorithm.

Procedure

1. Import the OpenCV (cv2) library.
2. Read the input image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Convert the image to grayscale.
5. Apply the Canny edge detection function.
6. Display the detected edges.
7. Save the output image.

Code

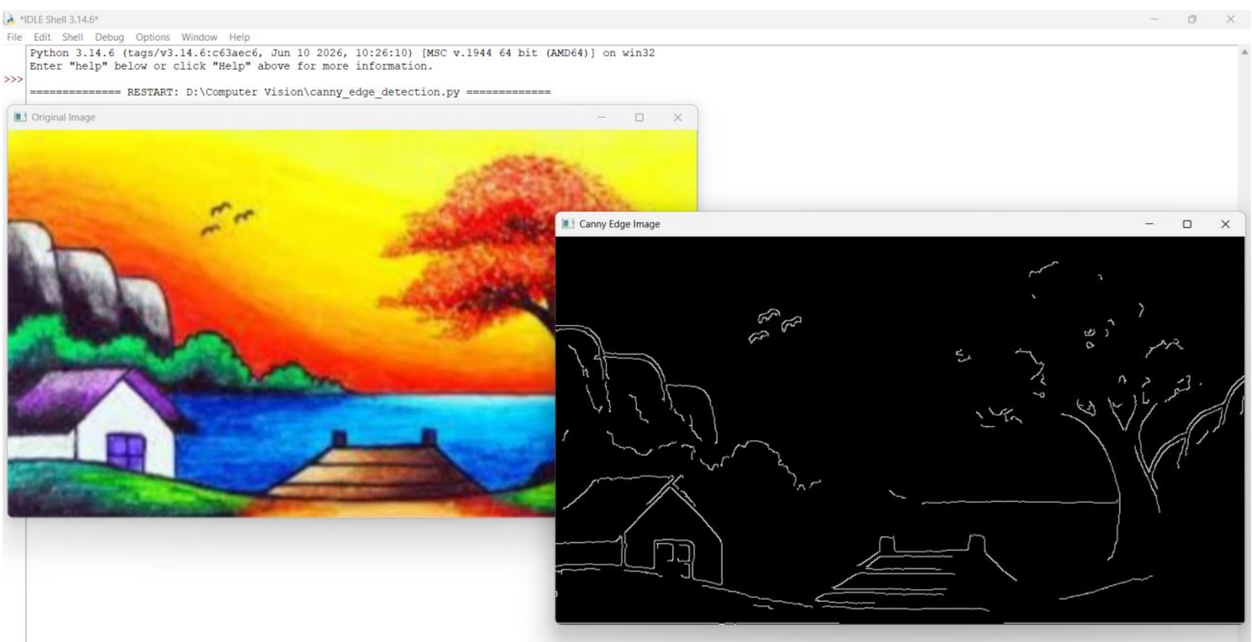
```
import cv2


# Read image
image = cv2.imread("nature.jpg")


# Check whether image is loaded
if image is None:
   print("Image not found!")
else:
   # Display original image
   cv2.imshow("Original Image", image)


   # Convert image to grayscale
   gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)


   # Apply Canny Edge Detection
   edges = cv2.Canny(gray, 100, 200)


   # Display edge image
   cv2.imshow("Canny Edge Image", edges)


   # Save edge image
   cv2.imwrite("canny_output.jpg", edges)
```

Input

Input Image: nature.jpg

Output

1. **Original Image**
    - Displays the original color image.
2. **Canny Edge Image**
    - Displays the outline (edges) of the image.
3. **Saved File**
    - canny_output.jpg is created in the project folder.



Result

The image was successfully read using OpenCV, the outlines were detected using the Canny edge detection algorithm, and the output image was displayed and saved successfully.